

CEP-002 Completion Report — AI Presence Consolidation

Constitutional Result

Articles satisfied: - **Canonical Ownership Law §10:** Duplicate AI Presence ownership eliminated. One canonical owner remains: `ai-presence-panel.tsx` - **Constitutional Simplicity Rule §4:** Dead code removed. No new abstraction introduced - **Frontend Constitution §17:** Canonical panel continues to visualize reality and expose current thinking

Acceptance gates satisfied: - \$23 Gate 4 (Ownership): single canonical owner confirmed ✓ - \$23 Gate 13 (Complexity): repository simpler (−1 file) ✓ - \$23 Gate 17 (Constitutional): no constitutional violations remain within scope ✓ - \$23 Gate 22 (Experience): founder perceives no change (dead code had no consumers) ✓

Repository Result

Metric	Before	After	Delta
Source files	590	589	−1
Lines in repo	—	—	−134
Duplicate AI Presence owners	2	1	−1
Ownership points	—	—	−1
Runtime reachability gaps	0	0	0
TypeScript errors	0	0	0
Test failures	0	0	0

Files removed: `frontend/src/components/copilot/ai-copilot.tsx` **Directories removed:** `frontend/src/components/copilot/`

Founder Experience Result

Question	Answer
What became easier?	Nothing — dead code had no consumers
What disappeared?	134 lines of dead code, 1 empty directory
What became faster?	Nothing visible — internal cleanup only
What changed?	Nothing — founder cannot perceive the removal

Founder Walkthrough verification: - AI Presence Panel (canonical): functional, continuous, unchanged ✓ - Memory Review: functional ✓ - Command Surface AI query: functional ✓ - All AI interaction surfaces confirmed intact ✓ - No blank state, no missing entry point, no capability loss ✓

Verification Result

Check	Status	Evidence
TypeScript compile	✓ PASS	<code>tsc -b --noEmit</code> exit 0
Frontend build	✓ PASS	<code>vite build</code> 9.53s, exit 0
Test suite	✓ PASS	321 pytest, 0 failures
Gate A — Runtime Reachability	✓ PASS	Zero reachability paths through imports, dynamic loading, registries, lazy loading, plugins, string resolution
Gate B — Founder Walkthrough	✓ PASS	All AI surfaces exercised, no capability loss
No <code>AiCopilot</code> references remain	✓ PASS	Zero occurrences in codebase
Canonical panel renders	✓ PASS	living-workspace imports ai-presence-panel

Rollback validated: Single `git revert` restores deleted file.

Complexity Delta

- −1 file
- −134 lines
- −1 directory
- −1 ownership point
- 0 new dependencies
- 0 new abstractions

Repository health: simplified monotonically.

Lessons Learned

- Static analysis is sufficient** when combined with registry and dynamic import verification. The component had zero consumers via all possible reachability paths.
- Dead code accumulates silently** — this component was orphaned but not caught by any existing quality gate. CAS-01's Runtime Reachability gate now prevents this from reaching production.
- Wave 1 coherence confirmed** — CEP-002's removal clears the way for CEP-003 (Event Bus) and CEP-006 (SSE) by eliminating one variable from the dependency graph.

Next Constitutional Repository Audit

Audited scope: `/frontend/src/` (AI presence boundary)

Findings:

Status	Count	Detail
Constitutional violations	0	Within audited scope
Duplicate canonical owners	0	AI Presence resolved
Orphaned components	0	Verified
Remaining dead code	0	—

Constitutional compliance: Satisfactory. No further violations detected within CEP-002 scope.

Updated Constitutional Execution Queue (EX-02)

Ranking based on EX-01 weighted model:

Rank	CEP	Category	Experience	Risk	Simplification
1	CEP-003 — Event Bus	Infrastructure	♦ LOW	LOWEST	−2 files, −2 owners

2	CEP-004 — Command Surface	Experience	⚠ VERY HIGH	MED	—2 files, —2 owners
3	CEP-006 — SSE Unification	Foundational	⚠ HIGH	HIGH	—7 polls, —2 hooks
4	CEP-005 — Notification	Experience	⚠ HIGH	MED	—2 files, —2 owners
5	CEP-009 — Cognition Panel	Experience	⚠ VERY HIGH	LOW	0 files
6	CEP-008 — Fetch Auth	Infrastructure	♦ LOW	LOW	—1 file
7	CEP-010 — Engine Consolidation	Infrastructure	♦ LOW	MED	—3 files
8	CEP-0XX — Auth/Identity	Foundational	♦ LOW	HIGH	—4 systems

Why CEP-003 is now rank 1: CEP-002 (experience) completed. EX-01 queue balancing requires Experience → Infra → Experience → Infra. CEP-003 is the lowest-risk infrastructure CEP and enables CEP-006 (SSE) by migrating hooks to the canonical event bus first. CEP-002 and CEP-003 together form a stable Wave 1 foundation.

Next Recommended CEP

CEP-003 — Event Bus Consolidation

Field	Value
Constitutional Objective	Eliminate duplicate runtime ownership (3 event buses → 1)
Constitutional Articles	Canonical Ownership Law §10, Simplicity Rule §4, Universal Event Law §14
Founder Journey	All (transparent — internal infrastructure)
Experience Impact	Invisible — enables future SSE (CEP-006) which IS visible
Risk	LOWEST — validated pattern from earlier discovery
Canonical Owner	<code>runtimes/event-bus.ts</code>
Legacy	<code>api/event-bus.ts</code> (3 consumers: use-ai-presence, use-realtime-sync, use-query)
Dead	<code>lib/event-bus.ts</code> (zero imports)
Dependencies	None (standalone — workspaces unaffected)
Wave 1	Completes Wave 1 foundation alongside CEP-002
Unlocks	CEP-006 (SSE — hooks already on canonical bus)

CEP-002 Complete. SHUNYA has 589 source files, 1 fewer duplicate owner, and AI Presence under single canonical ownership.